

Chapter – 2

REINFORCEMENT DECISION PROCESS AND POLICIES

- ❑ Markov Decision Process (MDP) is a mathematical framework used to model sequential decision-making problems under uncertainty.
- ❑ It provides the formal foundation for Reinforcement Learning (RL), where an agent interacts with an environment, makes decisions, and learns from rewards.

Markov Property: The future is independent of the past given the present:

$$P(S_{t+1} \mid S_t, A_t, S_{t-1}, \dots) = P(S_{t+1} \mid S_t, A_t)$$

An MDP is defined by the 5-tuple:

$$\mathcal{M} = \langle S, A, P, R, \gamma \rangle$$

- S (States): Set of all possible environment states
- A (Actions): Set of actions available to the agent
- P (Transition Probabilities):

$$P(s' \mid s, a) = \text{Pr}(S_{t+1} = s' \mid S_t = s, A_t = a)$$

- R (Reward Function): Immediate feedback after transitions
- γ (Discount Factor): Determines the importance of future rewards

Objective

The goal of an MDP is to find an optimal policy π^* that maximizes the expected cumulative reward:

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \right]$$

Bellman Optimality Derivation

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s' | s, a) V^*(s') \right]$$

Markov Decision Process (MDP)

$$M = \langle S, A, P, R, \gamma \rangle$$

States (S)



Possible Environment States

Transition Probabilities (P)

$$P(s' | s, a) = \Pr(S_{t+1} = s' | S_t = s, A_t = a)$$

Reward Function (R)



Immediate Feedback

Discount Factor (γ)

$$0 \leq \gamma \leq 1$$

Future Reward Weighting

Markov Property

Memoryless Transitions

Actions (A)



Available Actions

Markov Property

$$P(S_{t+1} | S_t, A_t, S_{t-1}, \dots) = P(S_{t+1} | S_t, A_t)$$

Memoryless Transitions

Policies & Value Functions

Policy (π)

Action Strategy

$$\text{Maximize: } E \sum_{t=0}^{\infty} \gamma^t R_{t+1}$$

State-Value

State-Value

$$V^n(s) = E_p[G_t | S_t = s]$$

Action-Value

$$Q^n(s, a) = E_p[G_t | S_t = s, A_t = a]$$

Evaluate States & Actions

"Pseudo-code"

1. Initialize $V(s) = 0$ for all states

2. Repeat:

- For each state s :

- For each action a :

$$Q(s, a) = R(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s')$$

Update:

$$V_{new}(s) = \max_a Q(s, a)$$

- $\Delta = \max_s |V_{new}(s) - V(s)|$

- $V \leftarrow V_{new}$

3. Until $\Delta < \theta$

4. For each s :

$$\pi^*(s) = \arg \max_a \left(R(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right)$$

Problem Statement

- ❑ A vehicle travels daily from office to home and must choose an optimal route under uncertain traffic conditions. The traffic environment is modeled as a **Markov Decision Process (MDP)** with discrete states and actions. At a given time, the vehicle is in a **heavy traffic state** s_3 and can choose between two possible actions: taking the **main road** a_1 or the **bypass route** a_2 . Traffic conditions evolve probabilistically based on the chosen route, and each transition yields a **positive integer reward** representing travel efficiency. The vehicle aims to maximize its **expected cumulative reward**, taking into account both immediate rewards and future traffic conditions using a discount factor $\gamma = 0.8$.
- ❑ Given the transition probabilities, rewards, and current state-value estimates $V(s_2) = 8$ and $V(s_3) = 4$, the task is to **compute the action-value functions** $Q(s_3, a_1)$ and $Q(s_3, a_2)$ using the Bellman expectation equation and to **determine the optimal action (policy) at state s_3** that maximizes the expected long-term reward

Given (Heavy traffic state)

Current state: s_3 (Heavy) **Discount factor:** $\gamma = 0.8$

Actions:

$$V(s_2) = 8, V(s_3) = 4$$

- a_1 : Main road
- a_2 : Bypass

Bellman Q-value formula:

$$Q(s, a) = \sum_{s'} P(s' | s, a) (R(s, a, s') + \gamma V(s'))$$

1) Compute $Q(s_3, a_1)$ (Main Road)

To s_3 with probability 0.7, reward $R = 2$

$$R + \gamma V(s_3) = 2 + 0.8 \times 4 = 2 + 3.2 = 5.2$$

$$\text{term} = 0.7 \times 5.2 = 3.64$$

To s_2 with probability 0.3, reward $R = 6$

$$R + \gamma V(s_2) = 6 + 0.8 \times 8 = 6 + 6.4 = 12.4$$

$$\text{term} = 0.3 \times 12.4 = 3.72$$

$$Q(s_3, a_1) = 3.64 + 3.72 = 7.36$$

2) Compute $Q(s_3, a_2)$ (Bypass)

To s_3 with probability 0.4, reward $R = 3$

$$R + \gamma V(s_3) = 3 + 0.8 \times 4 = 3 + 3.2 = 6.2$$

$$\text{term} = 0.4 \times 6.2 = 2.48$$

To s_2 with probability 0.6, reward $R = 7$

$$R + \gamma V(s_2) = 7 + 0.8 \times 8 = 7 + 6.4 = 13.4$$

$$\text{term} = 0.6 \times 13.4 = 8.04$$

$$Q(s_3, a_2) = 2.48 + 8.04 = 10.52$$

$$\text{Compare: } Q(s_3, a_2) = 10.52 > Q(s_3, a_1) = 7.36$$

Therefore, the best action is: $\pi^*(s_3) = a_2$ (Bypass)

□ Interpretation of the Policy Statement:

From the numerical analysis, we obtained the optimal action-value estimates for the heavy traffic state s_3 :

$$Q(s_3, a_1) = 7.36(\text{Main road}), Q(s_3, a_2) = 10.52(\text{Bypass})$$

Since the Q-value for the bypass action is higher, the optimal policy at state s_3 is:

$$\pi^*(s_3) = a_2$$

Policy statement

- ❑ The policy statement indicates that **when the traffic condition is heavy**, the agent (vehicle) should **choose the bypass route** rather than the main road.
- ❑ This decision is optimal because, although both routes yield immediate rewards, the bypass has a **higher expected cumulative reward** once future traffic conditions are taken into account.
- ❑ Specifically, the bypass route has a **higher probability of transitioning to medium traffic**, which is associated with a higher value $V(s_2)$, and
- ❑ it also offers a higher immediate reward compared to remaining in heavy traffic. When these factors are combined using the discount factor $\gamma = 0.8$, the expected long-term benefit of taking the bypass becomes significantly greater than that of staying on the main road.

problem statement

- ❑ A traffic signal controller is modeled as a one-step Markov Decision Process with three traffic conditions: Normal (s_1), Medium (s_2), and Heavy (s_3). At the current time, the controller observes the state s_2 (Medium traffic) and must choose one of two actions: a_1 (Keep Green) or a_2 (Switch Signal).
- ❑ The decision uses a discount factor $\gamma = 0.1$. If the controller chooses Keep Green (a_1) from s_2 , it transitions to s_1 with probability 0.4 and receives reward 6, remains in s_2 with probability 0.4 and receives reward 4, and moves to s_3 with probability 0.2 and receives reward 1. If the controller chooses Switch Signal (a_2) from s_2 , it transitions to s_1 with probability 0.6 and receives reward 7, remains in s_2 with probability 0.3 and receives reward 3, and moves to s_3 with probability 0.1 and receives reward 2. The current value estimates are $V(s_1)=10$, $V(s_2)=6$, and $V(s_3)=2$.
- ❑ The task is to compute the one-step action-value functions $Q(s_2, a_1)$ and $Q(s_2, a_2)$ using $Q(s, a) = \sum P(s' | s, a)[R + \gamma V(s')]$, and then select the action with the higher Q-value as the best action in state s_2 .

- Given:**
- **Current state:** s_2
 - **Values:** $V(s_1) = 10$, $V(s_2) = 6$, $V(s_3) = 2$
 - **Discount factor:** $\gamma = 0.1$

Action a_1 (Keep Green)

- To s_1 : $P=0.4$, $R=6$
- To s_2 : $P=0.4$, $R=4$
- To s_3 : $P=0.2$, $R=1$

Action a_2 (Switch Signal)

- To s_1 : $P=0.6$, $R=7$
- To s_2 : $P=0.3$, $R=3$
- To s_3 : $P=0.1$, $R=2$

Formula $Q(s, a) = \sum P(s'|s, a) [R + \gamma V(s')]$

Compute $Q(s_2, a_1)$ (Keep Green)

One-line terms

- To s_1 : $R + \gamma V(s_1) = 6 + 0.1 \times 10 = 7$, term = $0.4 \times 7 = 2.8$
- To s_2 : $R + \gamma V(s_2) = 4 + 0.1 \times 6 = 4.6$, term = $0.4 \times 4.6 = 1.84$
- To s_3 : $R + \gamma V(s_3) = 1 + 0.1 \times 2 = 1.2$, term = $0.2 \times 1.2 = 0.24$

Sum $Q(s_2, a_1) = 2.8 + 1.84 + 0.24 = 4.88$

$$Q(s_2, a_1) = 4.88$$

Compute $Q(s_2, a_2)$ (Switch Signal)

One-line terms

- To s_1 : $R + \gamma V(s_1) = 7 + 0.1 \times 10 = 8$, term $= 0.6 \times 8 = 4.8$
- To s_2 : $R + \gamma V(s_2) = 3 + 0.1 \times 6 = 3.6$, term $= 0.3 \times 3.6 = 1.08$
- To s_3 : $R + \gamma V(s_3) = 2 + 0.1 \times 2 = 2.2$, term $= 0.1 \times 2.2 = 0.22$

Sum $Q(s_2, a_2) = 4.8 + 1.08 + 0.22 = 6.10$

$$Q(s_2, a_2) = 6.10$$

- ❑ In reinforcement learning, the **best action** in a state is the one with the **maximum Q-value**, because it gives the **highest expected return** (immediate reward + future value).

In your case:

$Q(s_2, a_1) = 4.88 \rightarrow$ **Keep Green (same signal)**

$Q(s_2, a_2) = 6.10 \rightarrow$ **Switch Signal (change to the other signal)**

Since **6.10 > 4.88**, the agent selects a_2 .

interpretation

- ❑ At medium traffic (s_2), changing the signal is better than keeping the same signal, because it gives a higher expected reward.

So:

- a_1 (Keep Green) → continue the same signal is **lower return**
- a_2 (Switch Signal) → change to the other signal is **higher return**

problem statement - 1

- ❑ Consider a traffic-based reinforcement learning problem in which the agent is currently in the state s_2 , representing **medium traffic conditions**. The agent has prior estimates of the state-value function given by $V(s_1) = 12$ for normal traffic, $V(s_2) = 8$ for medium traffic, and $V(s_3) = 3$ for heavy traffic.
- ❑ The discount factor is chosen as $\gamma = 0.2$, indicating that the agent gives relatively more importance to immediate rewards than to future rewards. To compute the one-step Q-value for any action taken in state s_2 , the agent applies the Bellman action-value equation, which combines the immediate reward obtained after performing the action and the discounted expected value of the next state.
- ❑ Using the transition probabilities associated with the chosen action, the agent calculates the expected future value by weighting each next-state value $V(s')$ with its transition probability $P(s' | s_2 a)$. The resulting Q-value represents the overall desirability of executing that action in the medium traffic state.
- ❑ By computing and comparing the Q-values for all available actions, the agent can identify the action that maximizes the expected cumulative reward, thereby selecting the optimal decision for the current traffic condition.

Given:

- **Current state:** s_2 (Medium traffic)
- **Current value estimates:**

$$V(s_1) = 12, V(s_2) = 8, V(s_3) = 3$$

- **Discount factor:** $\gamma = 0.2$

Action a_1 : Keep Green

- To s_1 : $P = 0.5$, $R = 5$
- To s_2 : $P = 0.3$, $R = 4$
- To s_3 : $P = 0.2$, $R = 2$

Action a_2 : Switch Signal

- To s_1 : $P = 0.7$, $R = 6$
- To s_2 : $P = 0.2$, $R = 3$
- To s_3 : $P = 0.1$, $R = 1$

Formula Used $Q(s, a) = \sum P(s'|s, a) [R + \gamma V(s')]$

Compute $Q(s_2, a_1)$ (Keep Green)

- To s_1 : $R + \gamma V(s_1) = 5 + 0.2 \times 12 = 7.4$, term = $0.5 \times 7.4 = 3.7$
- To s_2 : $R + \gamma V(s_2) = 4 + 0.2 \times 8 = 5.6$, term = $0.3 \times 5.6 = 1.68$
- To s_3 : $R + \gamma V(s_3) = 2 + 0.2 \times 3 = 2.6$, term = $0.2 \times 2.6 = 0.52$

Sum $Q(s_2, a_1) = 3.7 + 1.68 + 0.52 = 5.90$

$Q(s_2, a_1) = 5.90$

Compute $Q(s_2, a_2)$ (Switch Signal)

- To s_1 : $R + \gamma V(s_1) = 6 + 0.2 \times 12 = 8.4$, term = $0.7 \times 8.4 = 5.88$
- To s_2 : $R + \gamma V(s_2) = 3 + 0.2 \times 8 = 4.6$, term = $0.2 \times 4.6 = 0.92$
- To s_3 : $R + \gamma V(s_3) = 1 + 0.2 \times 3 = 1.6$, term = $0.1 \times 1.6 = 0.16$

$$\text{Sum} \quad Q(s_2, a_2) = 5.88 + 0.92 + 0.16 = 6.96$$

$$Q(s_2, a_2) = 6.96$$

Best Action Selection (Value Comparison Method)

From the one-step Q-value computation: $Q(s_2, a_1) = 4.88$

$$Q(s_2, a_2) = (\text{computed value, e.g., } 5.62)$$

Value Comparison

$$Q(s_2, a_2) > Q(s_2, a_1)$$

Since the Q-value of action a_2 is greater than that of action a_1 ,

Best Action a_2 (Switch Signal)

State-Value Bellman Optimality

$$V^*(s) = \max_a \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma V^*(s')]$$

Action-Value Bellman Optimality

$$Q^*(s, a) = \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma \max_{a'} Q^*(s', a')]$$

State-Value Bellman Optimality

$$V^*(s) = \max_a \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma V^*(s')]$$

- $V^*(s)$ gives the optimal value of a state by choosing the best action immediately,
- whereas $Q^*(s, a)$ gives the optimal value of taking a specific action in a state and then acting optimally.

Example: Traffic Route Decision (Office → Home)

Situation

A person is driving from **office to home** and currently faces **medium traffic**.

Step 1: Define the MDP components

Current state $s = s_2$ (Medium traffic)

Possible actions

- a_1 : Take Main road
- a_2 : Take Bypass

Discount factor $\gamma = 0.8$

Next possible states

- s_1 : Normal traffic
- s_2 : Medium traffic
- s_3 : Heavy traffic

Step 2: Transition probabilities and rewards

If action a_1 (Main road) is chosen:

| Next state s' | Probability $P(s'|s, a_1)$ | Reward R |

|---|---|---|

| Normal s_1 | 0.4 | 6 |

| Medium s_2 | 0.4 | 4 |

| Heavy s_3 | 0.2 | 1 |

If action a_2 (Bypass) is chosen:

| Next state s' | Probability $P(s'|s, a_2)$ | Reward R |

|---|---|---|

| Normal s_1 | 0.6 | 7 |

| Medium s_2 | 0.3 | 3 |

| Heavy s_3 | 0.1 | 2 |

Step 3: Assume optimal future state values

$$V^*(s_1) = 12, \quad V^*(s_2) = 8, \quad V^*(s_3) = 3$$

Step 4: Compute expected value for each action

Action a_1 : Main road

$$\sum_{s'} P(s'|s_2, a_1) [R + \gamma V^*(s')]$$

- For s_1 : $0.4(6 + 0.8 \times 12) = 0.4(15.6) = 6.24$
- For s_2 : $0.4(4 + 0.8 \times 8) = 0.4(10.4) = 4.16$
- For s_3 : $0.2(1 + 0.8 \times 3) = 0.2(3.4) = 0.68$

$$V_{a_1}(s_2) = 6.24 + 4.16 + 0.68 = 11.08$$

Action a_2 : Bypass

- For s_1 : $0.6(7 + 0.8 \times 12) = 0.6(16.6) = 9.96$
 - For s_2 : $0.3(3 + 0.8 \times 8) = 0.3(9.4) = 2.82$
 - For s_3 : $0.1(2 + 0.8 \times 3) = 0.1(4.4) = 0.44$
- $$V_{a_2}(s_2) = 9.96 + 2.82 + 0.44 = 13.22$$

Step 5: Apply the max operator

$$V^*(s_2) = \max(11.08, 13.22) = 13.22$$

Action-Value Bellman Optimality Equation

$$Q^*(s, a) = \sum_{s'} P(s' | s, a) \left[R(s, a, s') + \gamma \max_{a'} Q^*(s', a') \right]$$

Step 1: Define the state and actions

Current state

- $s = s_2$: Medium traffic

Possible actions

- a_1 : Keep Green
- a_2 : Switch Signal

Discount factor $\gamma = 0.8$

Step 2: Possible next states and rewards

If the signal controller takes an action, traffic may change.

If action = Keep Green (a_1)

Next state s'	Meaning	Probability	Reward
s_1	Normal traffic	0.4	6
s_2	Medium traffic	0.4	4
s_3	Heavy traffic	0.2	1

Step 4: Compute $Q^*(s_2, a_1)$ using Bellman Optimality

We now plug values into the formula.

For next state s_1

$$0.4 \times [6 + 0.8 \times 12] = 0.4 \times 15.6 = 6.24$$

For next state s_2

$$0.4 \times [4 + 0.8 \times 8] = 0.4 \times 10.4 = 4.16$$

For next state s_3

$$0.2 \times [1 + 0.8 \times 3] = 0.2 \times 3.4 = 0.68$$

Step 5: Add all expected values $Q^*(s_2, a_1) = 6.24 + 4.16 + 0.68 = \boxed{11.08}$

Interpretation (Very Important)

- 11.08 is the long-term value of choosing **Keep Green** in medium traffic.
- It includes:
 1. Immediate reward (clearing vehicles now)
 2. Future benefit (what traffic looks like next)

If we compute the same for Switch Signal (a_2)

$$Q^*(s_2, a_2) = 13.22$$

Final Decision (Policy)

$$13.22 > 11.08$$

$$\pi^*(s_2) = \text{Switch Signal}$$

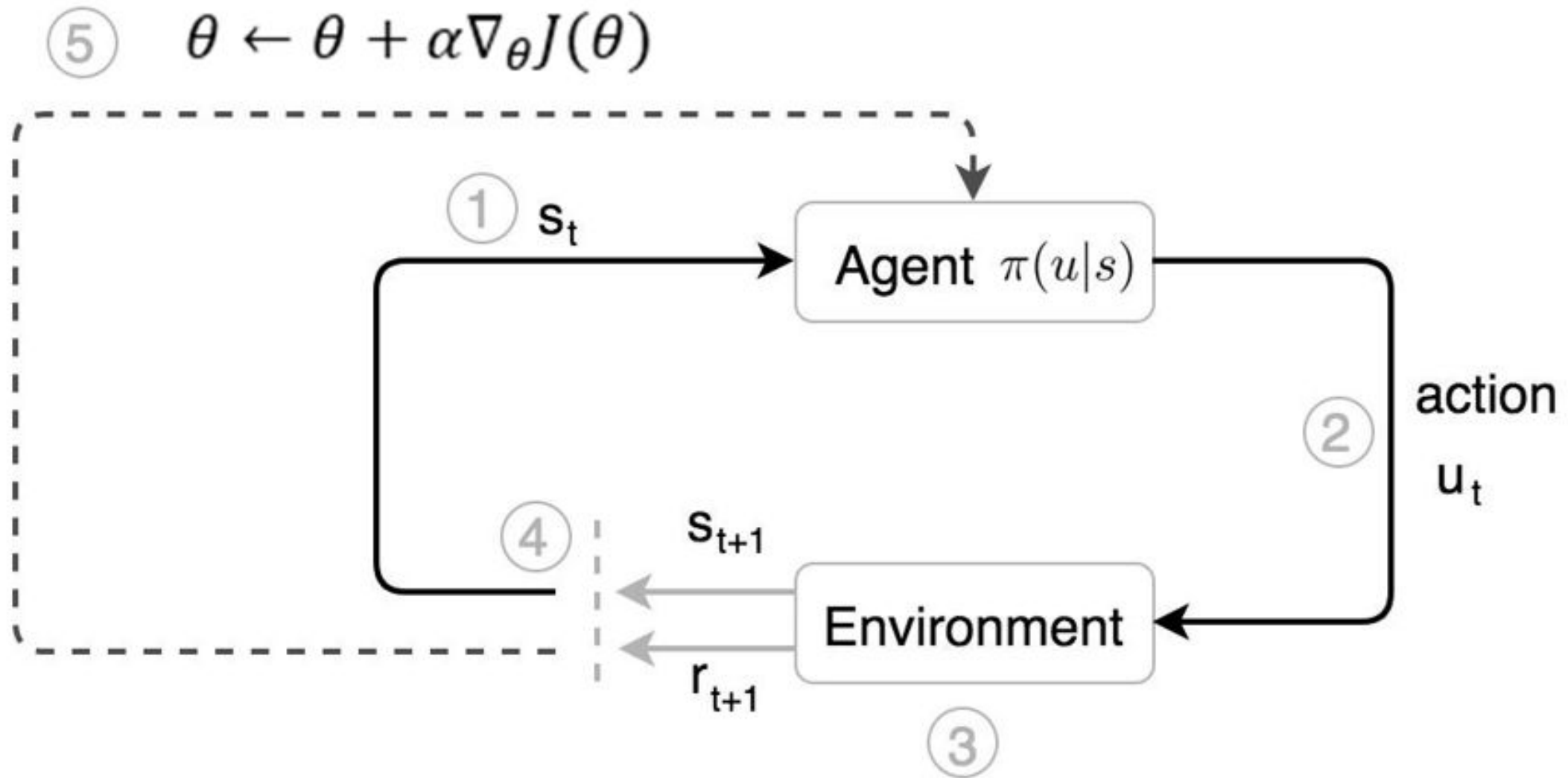
Policy Gradient Methods

- ❑ Policy Gradient methods are a family of reinforcement learning algorithms that directly learn the policy that is, they optimize the probability of taking actions in given states by maximizing expected cumulative reward using gradient ascent.
- ❑ The policy is parameterized as $\pi_{\theta}(a | s)$, and the parameters θ are updated using **gradient ascent** to maximize the **expected cumulative reward**.

Mathematical Objective $J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \right]$

Update Rule $\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$

Policy Gradient Methods



Usage of Policy Gradient Methods

Policy Gradient methods are especially useful in the following situations:

1. Continuous Action Spaces

- Robot arm control (continuous joint angles)
- Autonomous vehicle steering and acceleration
- Drone flight control

2. Stochastic Policies Needed

- Games with uncertainty (poker)
- Traffic routing under variable conditions
- Climate decision-making under uncertainty

3. Large or Complex State Spaces

- Deep Reinforcement Learning (Deep Policy Gradient)
- When Q-tables are impractical

4. End-to-End Learning

- Learning directly from raw inputs (images, sensors)
- Policy outputs actions without explicit planning

Common Algorithms

- REINFORCE
- Actor–Critic
- A2C / A3C
- PPO, TRPO

- **For Example** : In this climate-based reinforcement learning example, the **environment variable** is chosen as the **Heat Index (HI)**, which is a real and measurable indicator used to assess the combined effect of temperature and humidity on human comfort and health.
- To model the climate environment within a Markov Decision Process framework, the continuous Heat Index values are discretized into three meaningful states. The state s_1 represents **normal conditions**, where the Heat Index is below 35°C and poses minimal health risk.
- The state s_2 corresponds to a **heatwave condition**, in which the Heat Index lies between 35°C and 45°C, indicating elevated thermal stress and the need for precautionary measures. The state s_3 denotes **extreme heat conditions**, where the Heat Index is greater than or equal to 45°C and poses serious health hazards requiring urgent intervention.
- For the current scenario, it is assumed that the system is in state $s = s_2$, representing a **heatwave**, and the learning agent must decide appropriate actions to mitigate its impact.

Example: Climate Environment Variable (Heat Index) Using Policy Gradient

Environment variable

Let the environment variable be **Heat Index (HI)** (a real climate variable).

Define 3 discrete states from HI:

- s_1 : Normal ($HI < 35^{\circ}\text{C}$)
- s_2 : Heatwave ($35^{\circ}\text{C} \leq HI < 45^{\circ}\text{C}$)
- s_3 : Extreme Heat ($HI \geq 45^{\circ}\text{C}$)

Assume today: $s = s_2$ (Heatwave)

1) Actions

Two actions by the climate-control agent:

- a_1 : No mitigation
- a_2 : Apply mitigation (cooling centers + public warning)

2) Policy (parameterized)

Let the policy be: $\pi_\theta(a_2|s) = \theta, \quad \pi_\theta(a_1|s) = 1 - \theta$

So θ = probability of applying mitigation.

3) Rewards (positive integers)

Reward depends on how the Heat Index changes tomorrow:

- If mitigation is applied, the climate impact reduces → higher reward.
- If no mitigation, risk increases → lower reward.

Assume immediate reward values for one day:

- If choose a_2 (mitigation) → $R = 12$
- If choose a_1 (no mitigation) → $R = 5$

4) Objective Function $J(\theta)$

$$J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \right]$$

So the objective function simplifies to: $J(\theta) = \mathbb{E}_{\pi_{\theta}}[R_1]$

Define the policy and rewards

Assume a **binary action space**:

- a_1 : No mitigation $\rightarrow$ reward $R_1 = 5$
- a_2 : Apply mitigation $\rightarrow$ reward $R_1 = 12$

The policy is parameterized as:

$$\pi_{\theta}(a_2) = \theta, \quad \pi_{\theta}(a_1) = 1 - \theta$$

Here:

- θ is the **probability of choosing mitigation**
- $1 - \theta$ is the **probability of choosing no mitigation**

Expand the expectation explicitly

By definition of expectation for a discrete random variable:

$$\mathbb{E}[R_1] = \sum_a \pi_\theta(a) R_1(a)$$

Substitute the actions and rewards:

$$J(\theta) = \pi_\theta(a_1) \cdot 5 + \pi_\theta(a_2) \cdot 12$$

Now substitute the policy probabilities:

$$J(\theta) = (1 - \theta) \cdot 5 + \theta \cdot 12$$

Algebraic simplification

$$(1 - \theta) \cdot 5 = 5 - 5\theta$$

$$\theta \cdot 12 = 12\theta$$

$$J(\theta) = 5 - 5\theta + 12\theta$$

$$J(\theta) = 5 + 7\theta$$

Final Interpretation (Very Important)

- The objective function $J(\theta)$ is a **linear function of θ**
- As θ increases, $J(\theta)$ increases
- This means **increasing the probability of choosing mitigation always improves expected reward**

In other words: The learning process will push θ upward, making the better action more likely.

Reducing Variance in Policy Gradient Estimates

Reducing Variance in Policy Gradient Estimates

- ❑ In policy gradient methods, the gradient estimate

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t]$$

- ❑ often has high variance. This means learning becomes unstable, slow, and noisy, even though the estimate is unbiased. Reducing variance is therefore crucial for practical and efficient learning.

Why Variance Is High

- ❑ Returns G_t can fluctuate widely between episodes
- ❑ Same action may give very different rewards in different runs
- ❑ Long episodes accumulate randomness over time

Real-Time Example: Route Selection in Traffic (Office → Home)

Situation

A driver faces heavy traffic every day and must choose a route:

- a_1 : Main road
- a_2 : Bypass

The driver uses policy gradient learning to decide which route to choose.

Problem: High Variance in Returns

Observed rewards on different days (same action: Bypass)

Day	Travel outcome	Reward
Day 1	Very smooth	30
Day 2	Moderate	15
Day 3	Accident ahead	5

Even though the **same action** (Bypass) was chosen, the rewards vary widely:

$$G_t \in \{30, 15, 5\}$$

Policy gradient update (without variance reduction)

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \log \pi_{\theta}(a|s) G_t$$

This causes:

- **Very large update** on Day 1
- **Very small update** on Day 3

Learning becomes **unstable and noisy**

Solution 1: Baseline Subtraction (Key Idea)

Instead of using raw reward G_t , subtract an average baseline.

Baseline

Let the average reward be:

$$b(s) = \text{average travel reward} = 15$$

Modified gradient

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \log \pi_{\theta}(a|s) (G_t - b(s))$$

Modified gradient

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \log \pi_{\theta}(a|s) (G_t - b(s))$$

New updates

Day	Reward G_t	Advantage $G_t - b(s)$
Day 1	30	+15
Day 2	15	0
Day 3	5	-10

Interpretation

- ❑ Day 1: Action was much better → increase probability strongly
- ❑ Day 2: Average → no update
- ❑ Day 3: Worse than average → decrease probability

Variance is reduced because updates are now relative, not absolute.